Comparison across models

Methods

We compare two different likelihoods, Tweedie versus delta-glmm with Gamma likelihood, for models with fixed-effects and random effects for the spatial, temporal, and spatiotemporal process. Using glmmTMB, we define the spatial grid in 1km segments and assumed the exponential decay in correlation based on Euclidean distance. **Results**

- Spatial + temporal models provide the best.
- qqplot K-S tests signficant deviations from uniformity for random effects models.
- Comparison of training data.

mods	ziGamma	Tweedie
mean	947	1667
pos	426	752
pos + time	352	603
pos | time	796	1323
ENV + pos | time	246	521
ENV + pos + time	0	0
ENV	353	795
ENV + fixed.pos	351	756

The qqplots for the models with random effects are not great. For the ziGamma

Conversely, for the fixed effect model that residuals look better, but not great.

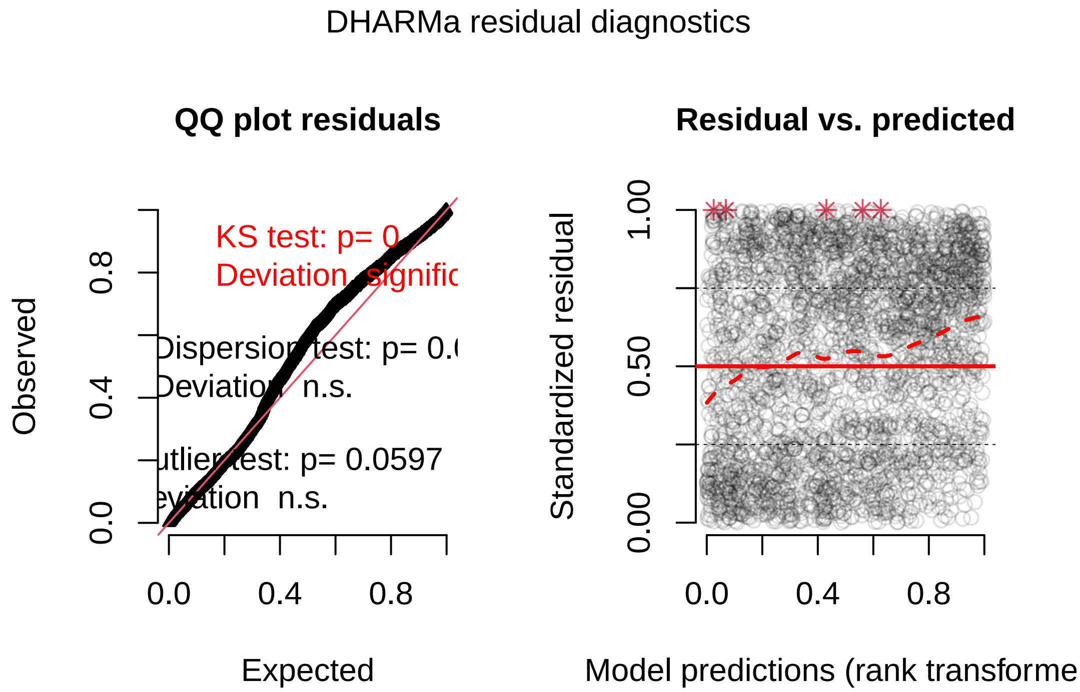

Figure 1: Standardized residuals for simulated data for zero inflated Gamma model.

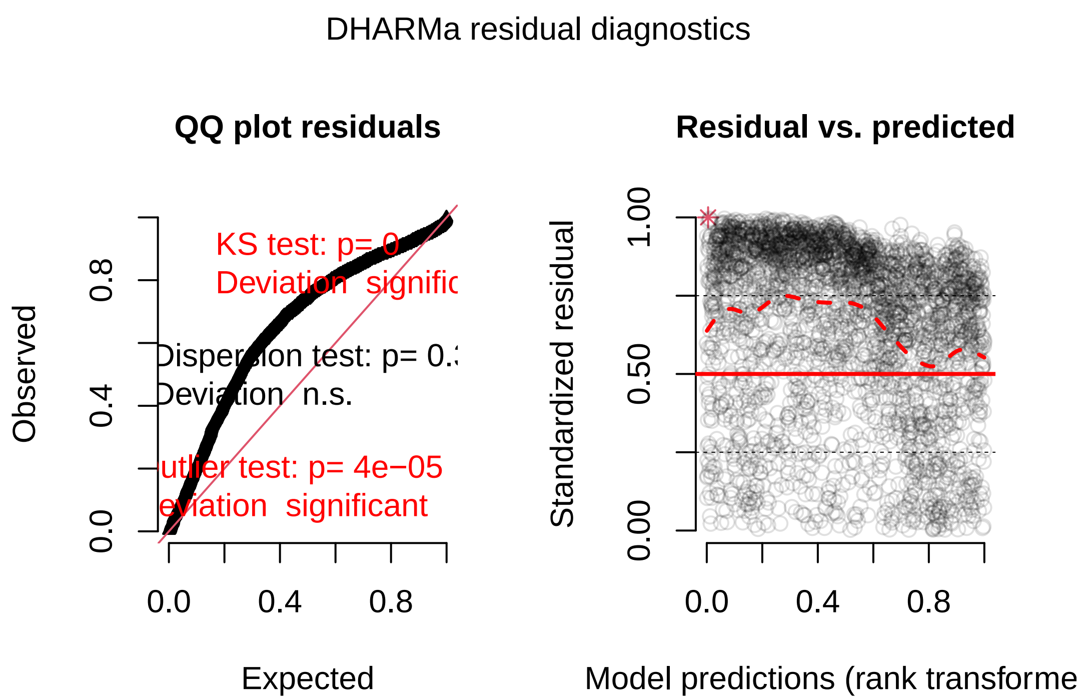

Figure 2: Standardized residuals for simulated data for Tweedie model.

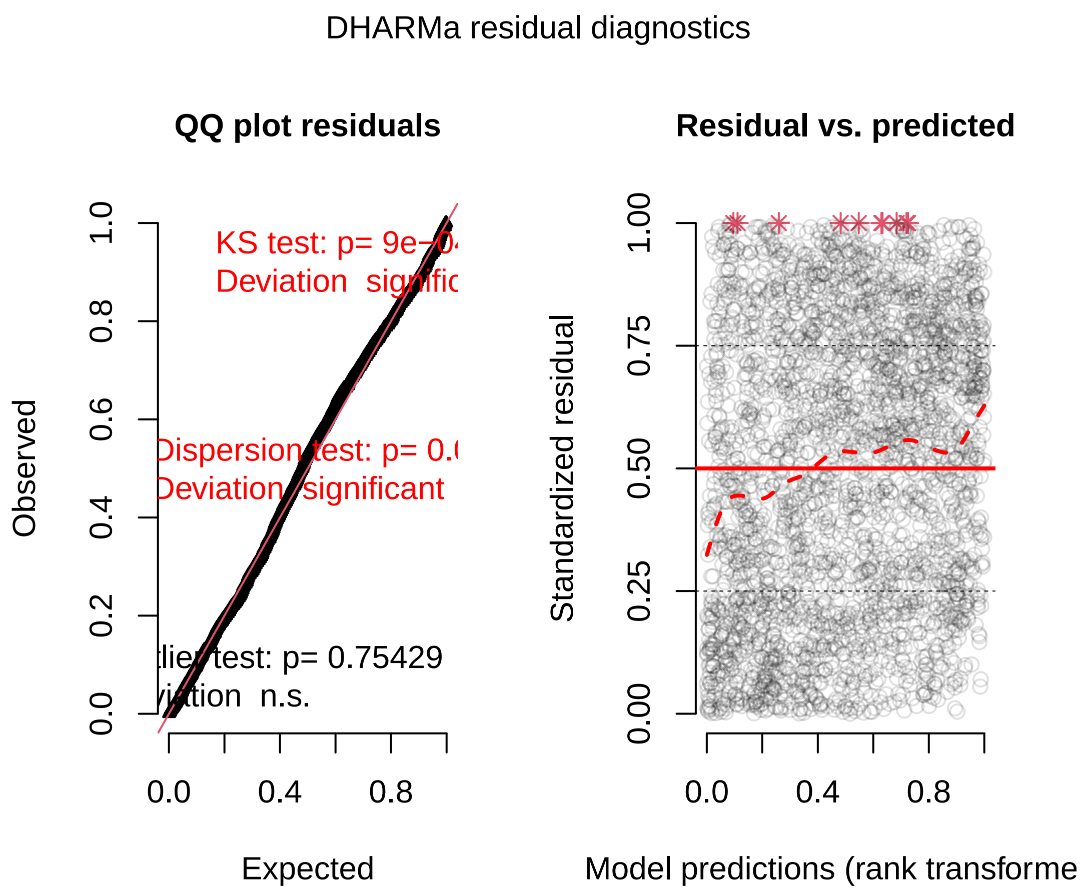

Figure 3: Standardized residuals for simulated data for zero inflated Gamma model.

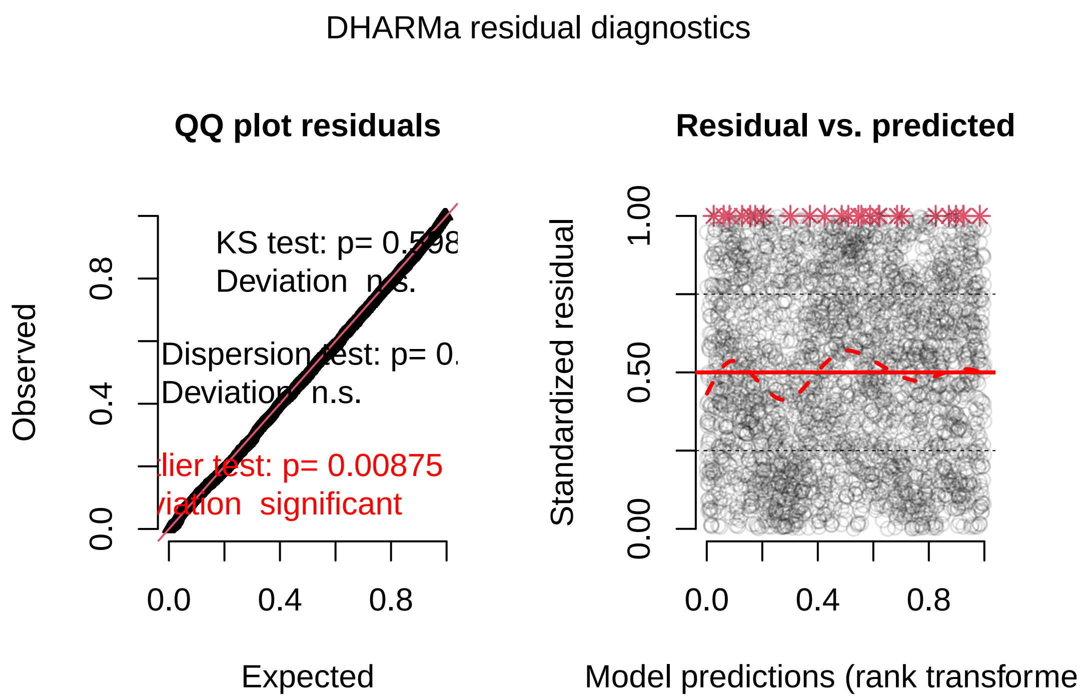

Figure 4: Standardized residuals for simulated data for Tweedie model.